Since Z_1 and Z_2 are independent,

$$\mathrm{Cov}(Z_1, Z_2) = E(Z_1 Z_2) - E(Z_1)E(Z_2) = 0 - 0 = 0$$

$$\implies \boxed{\mathrm{Corr}(Z_1, Z_2) = 0}$$

$$Y_1 = Z_1, \qquad Y_2 = \rho_{12} Z_1 + \sqrt{1 - \rho_{12}^2}\, Z_2$$

$$\begin{aligned} E(Y_2) &= \rho_{12} E(Z_1) + \sqrt{1 - \rho_{12}^2}\, E(Z_2) \\ &= 0 \end{aligned}$$

$$\implies \boxed{E(Y_2) = 0}$$

$$\begin{aligned} \mathrm{Var}(Y_2) &= \mathrm{Var}\left(\rho_{12} Z_1 + \sqrt{1 - \rho_{12}^2}\, Z_2\right) \\ &= \rho_{12}^2 \mathrm{Var}(Z_1) + (1 - \rho_{12}^2)\mathrm{Var}(Z_2) \end{aligned}$$

$$\because \ Z_1, Z_2 \text{ are independent}, \qquad \mathrm{Cov}(Z_1, Z_2) = 0$$

$$\mathrm{Var}(Y_2) = \rho_{12}^2 + (1 - \rho_{12}^2) = 1$$

$$\implies \boxed{\mathrm{Var}(Y_2) = 1}$$

$$\begin{aligned} \mathrm{Cov}(Y_1, Y_2) &= \mathrm{Cov}\left(Z_1, \rho_{12} Z_1 + \sqrt{1 - \rho_{12}^2}\, Z_2\right) \\ &= \rho_{12} \mathrm{Cov}(Z_1, Z_1) + \sqrt{1 - \rho_{12}^2}\, \mathrm{Cov}(Z_1, Z_2) \\ &= \rho_{12} \end{aligned}$$

$$\therefore \ \mathrm{Corr}(Y_1, Y_2) = \frac{\mathrm{Cov}(Y_1, Y_2)}{\sqrt{\mathrm{Var}(Y_1)\mathrm{Var}(Y_2)}} = \frac{\rho_{12}}{\sqrt{1 \cdot 1}} = \rho_{12}$$

$$\implies \boxed{\mathrm{Corr}(Y_1, Y_2) = \rho_{12}}$$

Now let

$$Y_3 = aZ_1 + bZ_2 + cZ_3, \qquad c \geq 0$$

Since $E(Z_1) = E(Z_2) = E(Z_3) = 0$,

$$E(Y_3) = aE(Z_1) + bE(Z_2) + cE(Z_3) = 0$$

which is already satisfied.

$$\begin{aligned}\mathrm{Var}(Y_3) &= \mathrm{Var}(aZ_1 + bZ_2 + cZ_3) \\ &= a^2 + b^2 + c^2\end{aligned}$$

$$\because \ Z_1, Z_2, Z_3 \text{ are independent}$$

Given $\mathrm{Var}(Y_3) = 1$,

$$a^2 + b^2 + c^2 = 1$$

Also,

$$\begin{aligned}\mathrm{Corr}(Y_1, Y_3) &= \frac{\mathrm{Cov}(Y_1, Y_3)}{\sqrt{\mathrm{Var}(Y_1)\mathrm{Var}(Y_3)}} \\ &= \mathrm{Cov}(Y_1, Y_3)\end{aligned}$$

since $\mathrm{Var}(Y_1) = \mathrm{Var}(Y_3) = 1$.

$$\begin{aligned}\rho_{13} &= \mathrm{Cov}(Z_1, aZ_1 + bZ_2 + cZ_3) \\ &= a\end{aligned}$$

$$\implies \boxed{a = \rho_{13}}$$

Next,

$$\begin{aligned}\rho_{23} &= \mathrm{Corr}(Y_2, Y_3) \\ &= \mathrm{Cov}(Y_2, Y_3)\end{aligned}$$

since $\mathrm{Var}(Y_2) = \mathrm{Var}(Y_3) = 1$.

$$\begin{aligned}\rho_{23} &= \mathrm{Cov}\left(\rho_{12}Z_1 + \sqrt{1-\rho_{12}^2}\, Z_2,\ aZ_1 + bZ_2 + cZ_3\right) \\ &= \rho_{12}a + \sqrt{1-\rho_{12}^2}\, b\end{aligned}$$

Substituting $a = \rho_{13}$,

$$\rho_{23} = \rho_{12}\rho_{13} + \sqrt{1-\rho_{12}^2}\, b$$

$$\sqrt{1-\rho_{12}^2}\, b = \rho_{23} - \rho_{12}\rho_{13}$$

$$\implies \boxed{b = \frac{\rho_{23} - \rho_{12}\rho_{13}}{\sqrt{1-\rho_{12}^2}}}$$

Now

$$a^2 + b^2 + c^2 = 1$$

so

$$c^2 = 1 - a^2 - b^2$$

$$c^2 = 1 - \rho_{13}^2 - \frac{(\rho_{23} - \rho_{12}\rho_{13})^2}{1 - \rho_{12}^2}$$

Since $c \geq 0$,

$$\implies \boxed{c = \sqrt{1 - \rho_{13}^2 - \frac{(\rho_{23} - \rho_{12}\rho_{13})^2}{1 - \rho_{12}^2}}}$$

Given constants μ_i and σ_i, define

$$X_1 = \mu_1 + \sigma_1 Y_1, \qquad X_2 = \mu_2 + \sigma_2 Y_2, \qquad X_3 = \mu_3 + \sigma_3 Y_3$$

Then

$$\begin{aligned} E(X_i) &= E(\mu_i + \sigma_i Y_i) \\ &= \mu_i + \sigma_i E(Y_i) \\ &= \mu_i \end{aligned}$$

$$\implies \boxed{E(X_i) = \mu_i}$$

Also,

$$\begin{aligned} \mathrm{Var}(X_i) &= \mathrm{Var}(\mu_i + \sigma_i Y_i) \\ &= \sigma_i^2 \mathrm{Var}(Y_i) \\ &= \sigma_i^2 \end{aligned}$$

$$\implies \boxed{\mathrm{Var}(X_i) = \sigma_i^2}$$

Finally,

$$\begin{aligned} \mathrm{Corr}(X_i, X_j) &= \frac{\mathrm{Cov}(X_i, X_j)}{\sqrt{\mathrm{Var}(X_i)\mathrm{Var}(X_j)}} \\ &= \frac{\sigma_i \sigma_j \mathrm{Cov}(Y_i, Y_j)}{\sigma_i \sigma_j} \\ &= \mathrm{Corr}(Y_i, Y_j) \\ &= \rho_{ij} \end{aligned}$$

$$\implies \boxed{X_i = \mu_i + \sigma_i Y_i \qquad (i = 1, 2, 3)}$$

This solution was submitted by **Jon**.

Published and refined by **Jon**.

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